

Exercise 6: Spatial Analytics

Three maps, three captions. No memo.

Item	Details
Points	100
Data	Exercise 6 starter workbook, or clean/mvu_student_locations.csv (1,400 rows: 33 home cities in 8 states; 3 campuses)
Reading	Learning Tableau 2025 (Milligan, 6th ed.), free through UTD Libraries: Ch 12
Walkthrough	Module 6 of the MVU Tableau Walkthrough follows these same steps, with checkpoints.
Submit	One packaged workbook (.twbx) with Worksheets 1–3 and a visible caption on each. No memo.

Set up: turn coordinates into geometry

1. Connect mvu_student_locations.csv (or open the starter). Check the four coordinate fields are Number (decimal), and set student_id to a Dimension.
2. Create Home Point = MAKEPOINT([home_latitude], [home_longitude]) and Campus Point = MAKEPOINT([campus_latitude], [campus_longitude]).

Checkpoint: How many rows are in the locations file?

Worksheet 1: density, and what it really shows

1. Drag Home Point onto the canvas; change the mark type to Density; adjust Intensity; use a plain basemap.
2. Drag Campus Point onto Add a Marks Layer. Set that layer to Shape, larger size, with campus_name on Label.

Worksheet 2: distance and service areas

1. Create Distance to Campus = DISTANCE([Home Point], [Campus Point], 'miles') and Within Service Area = IF [Distance to Campus] <= 50 THEN "Within 50 miles" ELSE "Beyond 50 miles" END.
2. Home Point on the canvas, Within Service Area on Color.
3. Create Service Area = BUFFER([Campus Point], 50, 'miles') and add it as a marks layer at about 25% opacity, underneath the points.
4. On a duplicate sheet, break the result down by campus_name.

Checkpoint: How many of the 1,400 students live within 50 miles of their campus?

Worksheet 3: origin-destination flows

1. Create Commute Line = MAKELINE([Home Point], [Campus Point]). Drag it onto the canvas.
2. student_id on Detail (so each line draws separately), campus_name on Color, opacity about 30%.
3. Filter to one campus at a time to see each catchment.

Fix all three maps

1. Retitle each map to say what it shows, not what it might be taken to show (“Where students live” beats “Recruitment density”).
2. On the buffer and flow maps, either filter out MVU Online or say plainly in the caption why it’s included.

Your three captions

Each worksheet carries a caption: a few sentences answering its question, written while the view is in front of you. Turn it on with Worksheet > Show Caption, then double-click below the view to edit.

Sheet	What your caption must answer
Worksheet 1	What does the density map demonstrate, and what does it not? Could it tell you where MVU recruits well?
Worksheet 2	What share of students at each campus live within 50 miles? What does the Online result actually mean?
Worksheet 3	What do the flow lines represent for each campus? Are any of them journeys that don’t happen?

Before you submit: open each sheet and confirm the caption is showing. A caption toggled off still holds your text but won’t be visible when your instructor opens the workbook.

Submit for Exercise 6

- Worksheet 1: a density map of student homes with campus locations as a second layer
- Worksheet 2: a service-area map using DISTANCE and BUFFER, with the within/beyond split visible
- Worksheet 3: a flow map built with MAKELINE
- A caption on each naming what the map does and does not demonstrate
- Packaged workbook (.twbx). No memo.